Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -1(Adv-2019-P1-Model)** **Date: 25-03-2026**

Time: 09.00Am to 12.00Pm **GTA-20** **Max. Marks: 186**

KEY SHEET

PHYSICS

1	**C**	2	**B**	3	**A**	4	**C**	5	**AB**	6	**BC**
7	**BCD**	8	**AC**	9	**AB**	10	**BC**	11	**ABCD**	12	**ABD**
13	**1**	14	**8**	15	**5**	16	**8**	17	**2**	18	**4**

CHEMISTRY

19	**A**	20	**C**	21	**D**	22	**C**	23	**ABCD**	24	**AB**
25	**BD**	26	**ABD**	27	**ABCD**	28	**D**	29	**CD**	30	**BC**
31	**8**	32	**0**	33	**5**	34	**141**	35	**10**	36	**99.1-99.2**

MATHEMATICS

37	**D**	38	**A**	39	**C**	40	**B**	41	**ABC**	42	**ABD**
43	**AD**	44	**ABD**	45	**ABC**	46	**BC**	47	**ABC**	48	**ABC**
49	**0**	50	**6**	51	**1**	52	**24**	53	**289**	54	**36**

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SOLUTIONS

PHYSICS

1. $dR = \frac{1}{6}\left(\frac{2h\ell d\ell + \ell^2 dh}{h^2}\right) + \frac{dh}{2}$

2. Apply ampere's circuital law.

3. Radiation pressure = $\frac{I}{C}(1+r)\cos^2\theta$

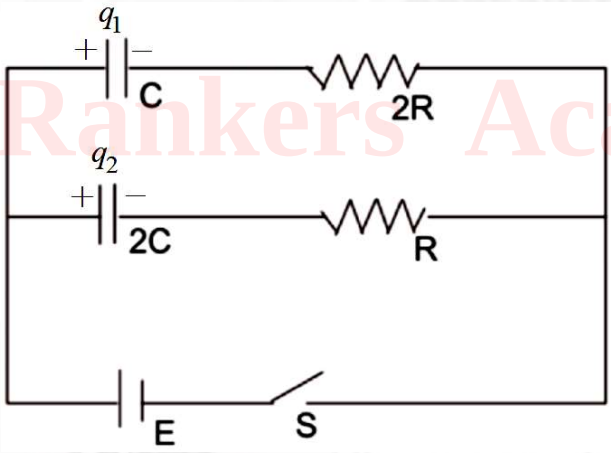
4. Conceptual

5. $V\frac{dn_1}{dt} = \frac{A}{4}n_0V - \frac{A}{2}n_1V \Rightarrow \int_{n_0}^{n_1}\frac{dn_1}{\frac{AV}{2}\left\{\frac{n_0}{2}-n_1\right\}} = \int_0^t \frac{dt}{V}$

$$\Rightarrow \ell n\left\{\frac{n_1 - \frac{n_0}{2}}{\frac{n_0}{2}}\right\} = \frac{-AVt}{2V} \Rightarrow n_1 = \frac{n_0}{2}\left\{1 + e^{\frac{-AVt}{2V}}\right\}$$

6. Force on $Q = \frac{PQ}{4\pi\varepsilon_0 r^3}$ is equal and opposite of force on dipole.

7.



$$E - \frac{q_1}{C} - 2R\frac{dq_1}{dt} = 0$$

$$E - \frac{q_2}{2C} - R\frac{dq_2}{dt} = 0$$

8. $B\ell v - L\frac{di}{dt} = 0$

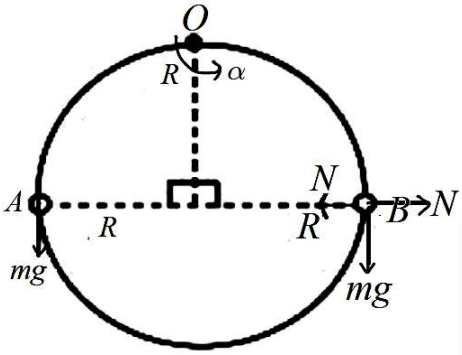
$$i\ell B = m\frac{dv}{dt}$$

9. $P = \rho g x + \rho v^2$

$$W = \int_0^L P dx = \frac{1}{2}\rho g L^2 + \rho v^2 L$$

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10. Line joining the rails is IAR.



11.

$$mgR - NR = 2mR^2\alpha$$

$$N = MR\alpha$$

12. Spring force doesn't changes immediately just after cutting string.

13. $$\frac{(T_1 - T_2)AK}{u} = 4eA\sigma T_0^3 (T_2 - T_0)$$

$$\Rightarrow (\Delta T_1 - \Delta T_2)\frac{AK}{u} = 4eA\sigma T_0^3 \Delta T_2 \Rightarrow \Delta T_1 - \Delta T_2 = \frac{4e\sigma \ell T_0^3}{K}\Delta T_2$$

$$\Rightarrow \Delta T_1 = \left\{1 + \frac{4e\sigma \ell T_0^3}{K}\right\}\Delta T_2$$

14. $$r = \frac{mv}{qB} = \frac{v}{\sigma B_0}$$

15. $$2(h - 720) + \frac{120^2 - 90^2}{20} = 330 \times 3$$

$$\Rightarrow 2(h - 720) + 315 = 990$$

$$\Rightarrow 2(h - 720) = 675 \Rightarrow h = 1057.5$$

16. $$m_1 = 2 \Rightarrow \frac{D - x_1}{x_1} = 2 \Rightarrow \frac{D}{x_1} - 1 = 2 \Rightarrow x_1 = \frac{D}{3} = 0.6m$$

$$x_2 = 1.2m$$

$$f = \frac{0.6 \times 1.2}{1.8} = 0.4m$$

17. $$V = \omega\sqrt{2Lx - x^2} = 10\sqrt{2 \times 1 \times 0.4 - 0.4^2}$$

$$= 10\sqrt{0.8 - 0.16} = 10 \times 0.8 = 8m/s$$

18. $$d\tau = (\eta \cdot 2\pi r dr)\frac{r\omega}{h} \cdot r$$

$$\Rightarrow \tau = \frac{\pi\eta\omega R^4}{2h} = K\omega$$

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CHEMISTRY

19. Obs. 1 $\Rightarrow$ Zone refining $\Rightarrow$ high-purity semiconductors (Ge, Si, Ga, In, B) $\Rightarrow$ Set I. Obs. 2 $\Rightarrow$ Vapour-phase refining (volatile, decomposable compound) $\Rightarrow$ Ni, Zr, Ti $\Rightarrow$ Set II. Obs. 3 $\Rightarrow$ Distillation (low boiling point metal) $\Rightarrow$ Zn, Hg $\Rightarrow$ Set III.

20. Compound (F) contains one benzylic $C-C\ell$ bond and one aryl $C-C\ell$ bond. Only the benzylic chloride reacts with $AgNO_3$ to form $AgC\ell$, while the aryl chloride does not. Hence, only one mole of $AgC\ell$ is formed, not two.

21. Effusion: $\left(nX/\sqrt{M}X\right):\left(nY/\sqrt{M}Y\right)=1:2\Rightarrow(nX/4):(nY/2)=1:2\Rightarrow nX=nY$.
Let extent reacted = ξ. Then nX = $3-\xi$ and nY = 2ξ. Using
$nX=nY:3-\xi=2\xi\Rightarrow\xi=1$.
So, at 10 s: nX = 2 mol, nY = 2 mol, nZ = 1 mol; total moles = 5.

22. Formation of red vapours with $K_2Cr_2O_7/conc.H_2SO_4$ confirms chromyl chloride $\left(CrO_2C\ell_2\right)$. Absorption in $NaOH$ gives chromate, which forms brick-red Ag_2CrO_4 with $AgNO_3$. Black precipitate with KI confirms BiI_3, which dissolves in excess KI forming an orange complex.

23. (A) $I_2+3C\ell_2\rightarrow 2IC\ell_3$ (inert solvent)
(B) HIO_3 oxidizes I^- to I_2 : $HIO_3+5HI\rightarrow 3I_2+3H_2O$
(C) $Ba(IO_3)_2+H_2SO_4\rightarrow BaSO_4\downarrow+2HIO_3$ (ppt drives)
(D) $IC\ell_3$ in $Ba(OH)_2$ gives $Ba(IO_3)_2$ with $BaC\ell_2, BaI_2$ by-products

24. The colourless gas (R) gives white fumes with $HC\ell$, identifying it as NH_3, indicating the presence of NH_4^+ in (P). The white precipitate (S) with $BaC\ell_2$ that is insoluble in $HC\ell$ confirms SO_4^{2-}. The green precipitate (Q) is $Fe(OH)_2$, which turns brown on exposure to air due to oxidation. Hence, statements (A) and (B) are correct.

25. On strong heating: KNO_3 decomposes to KNO_2 and O_2.
$Cu(NO_3)_2$ and $Pb(NO_3)_2$ decompose to the respective metal oxides, releasing NO_2 and O_2. NH_4NO_3 decomposes mainly to N_2O and H_2O.
NO_2 and O_2 are paramagnetic, whereas N_2O is diamagnetic. Hence, statement (C) is incorrect.

26. C is incorrect as C-H bond is weaker than N-H bond

27. $\Lambda\infty$ for the acid = $\Lambda\infty\left(H^+\right)+\lambda\infty\left(C\ell-CH_2-COO^-\right)=350+50$
$=400\Omega^{-1}cm^2mol^{-1}$. Degree of dissociation: $\alpha=\Lambda m/\Lambda\infty=100/400=0.25\Rightarrow(A)$ true. $\left[H^+\right]=\alpha C=0.25\times0.012=0.003M=3\times10^{-3}M\Rightarrow(B)$ true.
$Ka=\left(C\alpha^2\right)/(1-\alpha)=0.012\times(0.25)^2/(0.75)=0.001=10^{-3}$.
Thus, pKa = 3 $\Rightarrow$ (C) true.
$pH=-\ell og\left[H^+\right]=-\ell og\left(3\times10^{-3}\right)=3-\ell og3=3-0.48=2.52\Rightarrow(D)$ true.

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28. W: Grignard formation followed by hydrolysis gives the hydrocarbon (isobutane), not an alcohol $\Rightarrow$ (B) false.

X: Propyl chloride $\rightarrow$ Grignard $\rightarrow$ CO_2 $\rightarrow$ acid $\rightarrow$ $NaOH$ gives sodium butanoate; Kolbe electrolysis couples C_3H_7 radicals to form C_6H_{14} (hexane) $\Rightarrow$ (D) true.

Y: Grignard + CH_3CHO then PCC gives a methyl ketone (4-methyl-2-pentanone), which does not undergo Cannizzaro $\Rightarrow$ (A) false.

Z: $NaCN$ adds one carbon (nitrile), H_2 / Ni gives 1° amine; carbylamine gives isocyanide; $LiA\ell H_4$ reduces isocyanide to a 2° amine (N-methyl), not a 1° amine $\Rightarrow$ (C) false.

29. The first sequence (Kiliani–Fischer type) increases the carbon chain by one and yields an aldohexose framework as an epimeric mixture; hence writing a single fixed structure for M is not justified $\Rightarrow$ (A) incorrect. $PC\ell_5$ converts the five $-OH$ groups of M into $-C\ell$, giving a pentachloro derivative (pentachloride) $\Rightarrow$ (B) correct. $CH_3COC\ell$ /anhyd. $ZnC\ell_2$ does not simply give the pentaacetate under these conditions; chlorination (via $HC\ell$ generated) occurs at the anomeric center rather than forming only acetate esters $\Rightarrow$ (C) incorrect. Na–Hg(aq) reduces the aldehyde group of an aldohexose (M) to the corresponding hexitol; hence Q is sorbitol $\Rightarrow$ (D) correct.

30. NCERT

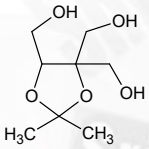
31. Moles of $C_2O_4^{2-}$ in 200 mL = $\left(4\times10^{-5}\right)/2=2\times10^{-5}mol$.

$\left[C_2O_4^{2-}\right]=\left(2\times10^{-5}\right)/(0.200)=1\times10^{-4}M$.

For $CaC_2O_4(s)\rightleftharpoons Ca^{2+}+C_2O_4^{2-}, Ksp=s^2=\left(10^{-4}\right)^2=10^{-8}$.

32. $v=Z^2RC\left(\frac{1}{n_i^2}-\frac{1}{n_f^2}\right), \frac{5}{9}=4\left(\frac{1}{4}-\frac{1}{n_f^2}\right)\Rightarrow n_f=3,$

$(n-\ell)_i=2-1=1, (n-\ell)_f=3-2=1. \left|(n-\ell)_f-(n-\ell)_i\right|=0$

33. 

34. $C_8H_{15}NO$

35. In the excited state $ns^2np\wedge(x-2)nd^2$, the central atom E has five unpaired electrons, hence it forms four bonds. Therefore, y = 5.Each fluorine atom has three lone pairs. Thus, lone pairs on fluorine atoms= $5\times3=15$. Total lone pairs in the molecule = 16, so lone pairs on E = 16 − 15 = 1.

36. One extraction: $\frac{n_{aq}}{n_0}=\frac{V_{aq}}{V_{aq}+K_DV_{org}}\Rightarrow\frac{n_{aq}}{n_0}=\frac{100}{100+4(100)}=\frac{1}{5}$

After 3 extractions: $\left(\frac{n_{aq}}{n_0}\right)3=\left(\frac{1}{5}\right)3=\frac{1}{125}$

Fraction in combined organic = $1-\frac{1}{125}=\frac{124}{125}$ % X in organic = $\frac{124}{124}\times100=99.2\%$

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MATHEMATICS

37. $P(x)=24x^{24}+23\left(x^{23}+x^{25}\right)+22\left(x^{22}+x^{26}\right)+.....+1\left(x^{13}+x^{47}\right)$

$=x+2x^2+.....+24x^{24}+.....+x^{47}$

$=x\left(1+x+.....+x^{23}\right)^2=\dfrac{x\left(x^{24}-1\right)^2}{(x-1)^2}$ $\qquad P(x)=0\Rightarrow x^{24}=1$

$x=0,\alpha,\alpha^2.....\alpha^{23}$ where $\alpha=e^{i2\pi/24}$

$\sum_{K=1}^{r}|b_K|=\sum_{K=1}^{r}\text{Im}\left(\left|x_K{}^2\right|\right)=8+4\sqrt{3}$

38. Let $L=\lim\limits_{x\to 0}\left(\int_0^1\left(by+a(1-y)\right)^x dy\right)^{\frac{1}{x}}$

Let $I=\int_0^1\left(by+a(1-y)\right)^x dy$

Let $by+a(1-y)=t,(b-a)dy=dt$ $\qquad\Rightarrow I=\int\dfrac{t^x}{(b-a)}dt=\dfrac{t^{x+1}}{(x+1)(b-a)}$

$I=\left[\dfrac{\left(by+a(1-y)\right)^{x+1}}{(x+1)(b-a)}\right]$ [Computer at $y=1$ and $y=0$]

$I=\dfrac{b^{x+1}-a^{x+1}}{(x+1)(b-a)}\Rightarrow L=\lim\limits_{x\to 0}\left(\dfrac{b^{x+1}-a^{x+1}}{(x+1)(b-a)}\right)^{\frac{1}{x}}$

$ln(L)=\lim\limits_{x\to 0}\dfrac{ln\left(\dfrac{\left(b^{x+1}-a^{x+1}\right)}{(x+1)(b-a)}\right)}{x}$

As x approaches 0, denominator and numerator approaches 0.
Hence, we can use L–hospital's rule.

$ln(L)=\lim\limits_{x\to 0}\left(\dfrac{b^{x+1}ln(b)-a^{x+1}ln(a)}{(x+1)(b-a)}-\dfrac{1}{x+1}\right)$

$ln(L)=\dfrac{1}{b-a}ln\dfrac{b^b}{a^a}-1\Rightarrow L=\dfrac{1}{e}\left(\dfrac{b^b}{a^a}\right)^{\frac{1}{(b-a)}}$

39. $a_0=\dfrac{17}{4}=4+\dfrac{1}{4}$ $\qquad a_1=a_0^2-2=4^2+\dfrac{1}{4^2}$

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$$a_2 = a_1^2 - 2 = 4^4 + \frac{1}{4^4} \quad a_n = a_{n-1}^2 - 2 = 4^{2^{n-1}} + \frac{1}{4^{2^{n-1}}}$$

$$\prod_{k=0}^{\infty}\left(1-\frac{1}{a_k}\right) = \lim_{x\to\infty}\left(1-\frac{1}{a_0}\right)\left(1-\frac{1}{a_1}\right).....\left(1-\frac{1}{a_n}\right)$$

$$= \lim_{n\to\infty} \frac{\left(4^2-4+1\right)\left(4^2-4^2+1\right).....\left(4^{2^n}-4^{2^{n-1}}+1\right)}{\left(4^2+1\right)\left(4^4+1\right).....\left(4^{2^n}+1\right)}$$

$$= \frac{4^2-1}{4^2+4+1}\lim_{n\to\infty}\frac{\left(4^{2^{n+1}}+4^{2^n}+1\right)}{\left(4^{2^{n+1}}-1\right)} = \frac{15}{21} = \frac{5}{7}$$

40. $$\ln f(x) = \lim_{n\to\infty}\frac{x}{n}\log\left\{\frac{n^{2n}\left(1+\frac{x}{n}\right)\left(\frac{1}{2}+\frac{x}{n}\right)...\left(\frac{1}{n}+\frac{x}{n}\right)}{n!n^{2n}\left(1+\frac{x^2}{n^2}\right)\left(\frac{1}{4}+\frac{x^2}{n^2}\right)...\left(\frac{1}{n^2}+\frac{x^2}{n^2}\right)}\right\}$$

$$= \lim_{n\to\infty}\frac{x}{n}\left[\sum_{r=1}^{n}\log\left(\frac{x}{n}+\frac{1}{r}\right) - \sum_{r=1}^{n}\log\left(\frac{rx^2}{n^2}+\frac{1}{r}\right)\right]$$

$$= \lim_{n\to\infty}\frac{x}{n}\left[\sum_{r=1}^{n}\log\left(1+\frac{rx}{n}\right) - \log r\right] - \sum_{r=1}^{n}\left(\log\left(1+\frac{r^2x^2}{n^2}\right) - \log r\right)$$

$$= \lim_{n\to\infty}\frac{x}{n}.\left[\sum_{r=1}^{n}\log\left(1+\frac{rx}{n}\right) - \sum_{r=1}^{n}\log\left(1+\frac{1^2x^2}{n^2}\right)\right] = x\int_0^1\left[\log(1+xy) - \log\left(1+x^2y^2\right)\right]dy$$

$$= \ln f(x) = \int_0^x \ln(1+t)\,dt - \int_0^x \ln\left(1+t^2\right)dt \Rightarrow \frac{f^1(x)}{f(x)} = \ln\left[\frac{1+x}{1+x^2}\right]$$

for $x>0,\ f(x)>0$ $\qquad \left(\frac{f'(x)}{f(x)}\right) > 0$ for $x\in(0,1)$, < 0 for $x\in(1,\infty)$

41. $$f(x)\,g\left(\frac{1}{x}\right) = \frac{1}{x} \quad \forall \quad x>0 \qquad \text{_____(1)}$$

$$f\left(\frac{1}{x}\right)g(x) = x \quad \forall \quad x>0 \qquad \text{_____(2)}$$

$g'(x) = f(x)$ [given] $\qquad \frac{1}{2} \Rightarrow \frac{f(x)}{f\left(\frac{1}{x}\right)}\frac{g\left(\frac{1}{x}\right)}{g(x)} = \frac{1}{x^2}$

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$$\Rightarrow \frac{g'(x)}{g'\left(\frac{1}{x}\right)}\frac{g\left(\frac{1}{x}\right)}{g(x)}=\frac{1}{x^2} \qquad \Rightarrow \frac{g'(x)}{g(x)}=\frac{1}{x^2}\frac{g'\left(\frac{1}{x}\right)}{g\left(\frac{1}{x}\right)}$$

$$\Rightarrow \quad \ln g(x)=-\ln g\left(\frac{1}{x}\right)+\ln c_1$$

$$\Rightarrow \quad g(x)g\left(\frac{1}{x}\right)=c_1 \qquad \text{_____(3)}$$

$$\frac{3}{1} \quad \Rightarrow \quad \frac{g(x)}{f(x)}=c_1x \quad \Rightarrow \quad \frac{g'(x)}{g(x)}=\frac{1}{c_1x}$$

$$l\ln g(x)=\frac{1}{c_1}\ln x+lnc_2 \qquad f(1)=2, \quad g'(1)=2$$

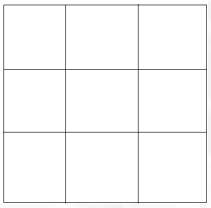
$$f(1)g(1)=1 \Rightarrow \quad g(1)=\frac{1}{f(1)}=\frac{1}{2} \quad \frac{g'(1)}{g(1)}=\frac{1}{c_1(1)}\Rightarrow c_1=\frac{g(1)}{g'(1)}=\frac{1}{4}$$

$$\ln g(x)=4\ln x+\ln\left(\frac{1}{2}\right)\Rightarrow g(x)=\left(\frac{x^4}{2}\right) \; f(x)=g'(x)=2x^3$$

42. $$S_k=\sum_{j=0}^{k}2^{k-j}\left({}^{k+j-1}C_j+{}^{k+j-1}C_{j-1}\right)=2.S_{k-1}+\frac{S_k}{2} \qquad \Rightarrow S_k=4S_{k-1}$$

We get $S_k=4^k \; \forall k\in W\left(S_0=1\right)$

43.



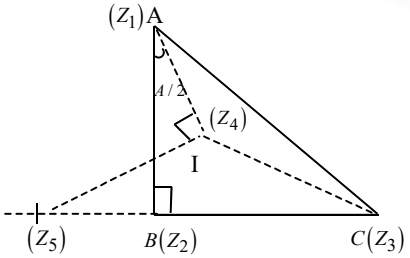
Case-1: If the centre square is coloured then two of the four corner cells have to be coloured as they do not share an common edge Number of ways = $4C_2=6$.

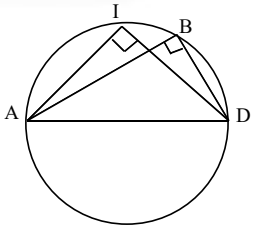
Case-2: If the centre square is not coloured Select any one of the eight squares and the remaining 2 squares can be selected in 6 ways but each possibility is over counted by a factor of 3.

Number of ways = $\frac{8*6}{3}=16$

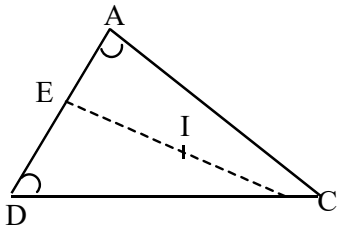
Total number of ways = 22

44.





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ADBI is a cyclic quadrilateral

$\angle AID = \angle ABD = \frac{\pi}{2}$

$\angle ADI = \angle ABI = \frac{\pi}{4}$ (chord AI subtends same angle on the circle)

$\Rightarrow \angle DAI = \frac{\pi}{4}$

AID is an isosceles triangle which is night angle at I

$AI = ID = \sqrt{2} \quad AD$

$\angle ADC = \angle ADI + \angle IDB = \frac{\pi}{4} + \angle DAI + \angle IAC = \angle DAC$

$\angle ADC \angle DAC \Rightarrow CD = CA$

$BD = CD - CB = CA - CB = b - a$

$AD^2 = AB^2 + DB^2 = c^2 + (b-a)^2 = c^2 + b^2 + a^2 - 2ab = 2b^2 - 2ab \quad = 2b\,(b-a)$

$AD = \sqrt{2}\sqrt{b(b-a)} \Rightarrow \frac{AD}{\sqrt{2}} = AI = DI$

CAD is isosceles triangle $\Rightarrow \angle CED = \frac{\pi}{2} \quad Arg\left(\frac{z_3 - z_1}{z_1 - z_5}\right) = \frac{-\pi}{2}$

45. A) $2\left(\tan^{-1}\pi - \tan^{-1}e\right) - ln\pi + 1 < 0$

Consider $f(x) = 2\left(\tan^{-1}x\right) - \ln x$

$f'(x) = \frac{2}{1+x^2} - \frac{1}{x} = \frac{2x - 1 - x^2}{x\left(1+x^2\right)} = \frac{-(x-1)^2}{x\left(x^2+1\right)} < 0$ for x > 0

$e < \pi \Rightarrow f(e) < f(\pi)$ $\quad 2\tan^{-1}(e) - \ln e > 2\tan^{-1}(\pi) - \ln(\pi)$

$0 > 2\left(\tan^{-1}(\pi) - \tan^{-1}(e)\right) - \ln\pi + 1$

Hence it is true

B) $\sum_{k=0}^{n} \frac{{}^nC_k}{n\left(2^n+1\right)} \le \sum_{k=0}^{n} \frac{{}^nC_k}{n.2^n + k} \le \sum_{k=0}^{n} \frac{{}^nC_k}{n.2^n}$

$\frac{{}^nC_k}{n.2^n + n} \le \frac{{}^nC_k}{n.2^n + k} \le \frac{{}^nC_k}{n.2^n + 0} \qquad 0 \le \sum_{k=0}^{\infty} \frac{{}^nC_k}{2^n + k} \le 0$

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C) Use $\sigma^2 = \dfrac{n_1\left(\sigma_1^2 + d_1^2\right) + n_2\left(\sigma_2^2 + d_2^2\right)}{n_1 + n_2}$

Where $d_1 = m_1 - a, d_2 = m_2 - a, a$ being the mean of the whole ground, Let m_2=mean of the second group

$15.6 = \dfrac{100\times 15 + 150\times m_2}{250} \Rightarrow m_2 = 16$

$\left(100\times 9 + 150\times\sigma^2\right)$

Thus, $13.44 = \dfrac{+100\times(0.6)^2 + 150\times(0.4)^2}{250} \qquad \Rightarrow \sigma = 4$

D) $\bar{x} = \dfrac{1}{101}\left[1 + (1+d) + (1+2d) + \ldots\ldots\ldots + (1+100d)\right]$

$= \dfrac{1}{101}\times\dfrac{101}{2}\left[1 + (1+100d)\right] = 1 + 50d\,.$

Mean deviation from mean $= \dfrac{1}{101}\left[\left|1 - (1+50d)\right| + |1 + d - \right]$

$(1+50d)\Big| + \ldots\ldots\ldots + \left|1 + 100d - (1+50d)\right|$

$255 = \dfrac{2|d|}{101}\{1 + 2 + \ldots\ldots\ldots + 50\} \qquad \Rightarrow \qquad 2\dfrac{|d|}{101}\dfrac{50(51)}{2} = \dfrac{2550}{101}|d|$

Now $\dfrac{2550}{101}|d| = 10.1$

46. $S(0,0), P(0,14), Q(0,-6) \qquad \dfrac{1}{SP} + \dfrac{1}{SQ} = \dfrac{2}{l} \qquad \Rightarrow l = \dfrac{42}{5}$

and $PQ = 4a\,Co\sec^2\theta$ *where* θ *is the angle made* PQ *with principle axis*

47. $f(x).f(y) + f(x+y) = e^x f(y) + e^y(x) + xy$

$\therefore f(x).f'(y) + f'(x+y) = e^x f'(y) + e^y f(x) + x$

Put y = 0

$f(x).f'(0) + f'(x) = e^x f'(0) + f(x) + x$

$\Rightarrow f'(x) - f(x) = x \Rightarrow f(x) = e^x - x - 1$

$\therefore (A)\lim\limits_{x\to 0}\dfrac{f(x)}{x^2} = \lim\limits_{x\to 0}\dfrac{e^x - x - 1}{x^2} = \dfrac{1}{2}.$

$(B)\int\limits_x^{x^2} tdt = \left(\dfrac{t^2}{2}\right)_x^{x^2} = \dfrac{x^4 - x^2}{2} > 0\forall|x| > 1$

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$(C)F(x)=f'(x)-f(x)=\left(e^x-1\right)-\left(e^x-x-1\right)=x$ which is increasing function so,

$F(x_2)>F(x_1)$ if $x_2>x_1$

(D) Now $f'(x)=e^x-1$ and $f''(x)=e^x>0\forall x\in R$

$\therefore f'(x)$ is increasing function or R

$\because f'(-1)=-ve$ and $f'(1)=+ve$ $\quad\therefore f'(c)=0$ has exactly are root in (-1 ,1).

i.e., one horizontal tangent. ln (-1,1)

48. $f(\alpha)=\lim\limits_{x\to 2}\left(\sin^x\alpha+\cos^x\alpha\right)^{\frac{1}{(x-2)}}\left(1^\infty\ form\right)$

$$=\begin{cases}\lim\limits_{e^{x\to 2}}\dfrac{\sin^x\alpha+\cos^x\alpha-1}{x-2} & ,\quad \alpha\in\left(0,\dfrac{\pi}{2}\right)\\ 1 & ,\quad \alpha=0,\dfrac{\pi}{2}\end{cases}$$

$$\lim_{e^{x\to 2}}\frac{\sin^x\alpha+\cos^x\alpha-\left(\sin^2\alpha+\cos^2\alpha\right)}{x-2}$$

$$=e^{\lim\limits_{x\to 2}}\frac{\sin^2\alpha\left(\sin^{x-2}\alpha-1\right)+\cos^2\alpha\left(\cos^{x-2}\alpha-1\right)}{x-2}$$

$$=e^{\sin^2\alpha\log_e\sin\alpha+\cos^2\alpha\log_e\cos\alpha}=e^{\log_e(\sin\alpha)^{\sin^2\alpha}+\log_e(\cos\alpha)^{\cos^2\alpha}}$$

$$=e^{\log_e(\sin\alpha)^{\sin^2\alpha}(\cos\alpha)^{\cos^2\alpha}}=(\sin\alpha)^{\sin^2\alpha}(\cos\alpha)^{\cos^2\alpha}$$

$$\therefore f(x)=\begin{cases}(\cos\alpha)^{\cos^2\alpha}.(\sin\alpha)^{\sin^2\alpha} & ,\quad \alpha\in\left(0,\dfrac{\pi}{2}\right)\\ 1 & ,\quad \alpha=0,\dfrac{\pi}{2}\end{cases}$$

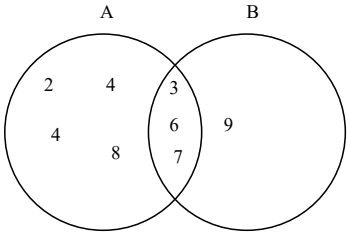
49. $\dfrac{1}{2}\cos^{-1}x=t$ $\qquad\dfrac{1+\tan t}{1-\tan t}+\dfrac{1-\tan t}{1+\tan t}=1$

$(1+\tan t)^2+(1-\tan t)^2=1-\tan^2 t$

$2+2\tan^2 t=1-\tan^2 t\quad\tan^2 t=\dfrac{-1}{3}$

No solution exists

50.



number of elements $=2^6-2^3$ $\qquad=64-8=56$

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51. $\dfrac{81^{x^2+y}+81^{y^2+z}+81^{z^2+x}}{3} \geq \left(81^{x^2+x+y^2+y+z^2+z}\right)^{1/3}$

$\dfrac{1}{3} \geq 3^{\left(x^2+x+y^2+y+z^2+z\right)\frac{4}{3}} \geq 3^{\left(\left(x+\frac{1}{2}\right)^2+\left(y+\frac{1}{2}\right)^2+\left(z+\frac{1}{2}\right)^2-\frac{3}{4}\right)\frac{4}{3}}$ $x=y=z=\dfrac{-1}{2}$

52. $A=\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

Let $x=\begin{bmatrix}1\\1\\1\end{bmatrix}$ $AX=\begin{bmatrix} a_{11}+a_{12}+a_{13} \\ a_{21}+a_{22}+a_{23} \\ a_{31}+a_{32}+a_{33} \end{bmatrix}=\begin{bmatrix}2\\2\\2\end{bmatrix}$

$\Rightarrow \quad AX=2X$

$A^2X=2AX=2(2X)=4X$

$A^3X=4AY=8X$

Let $A^3=\begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix}$

$A^3\begin{bmatrix}1\\1\\1\end{bmatrix}=\begin{bmatrix} x_1+x_2+x_3 \\ y_1+y_2+y_3 \\ z_1+z_2+z_3 \end{bmatrix}=\begin{bmatrix}8\\8\\8\end{bmatrix}$

Sum of all the element = 24

53. X : 2, 7, 12, ……., 10087, 10092

Y : 3, 10, 17, ……., 14122, 14129

$X\cap Y$:17, 52, 87, ……..

Let $m=n(X\cap Y)$

$\therefore 17+(m-1)\times 35 \leq 10127$

$\Rightarrow m=289$

54. $3\alpha^2+\beta=2013,\ 2\alpha^2+\gamma=2014 \Rightarrow 3\gamma-2\beta=2016$

$3(x-\alpha)^2+\beta=0 \Rightarrow \beta=-2(x-\alpha)^2=-3\lambda^2$ {$\because x$ in each case are different}

$2(x-\alpha)^2+\gamma=0 \Rightarrow \gamma=-2(x-\alpha)^2=-2\mu^2$

$6\lambda^2-6\mu^2=2016 \Rightarrow \lambda^2-\mu^2=336 \Rightarrow \lambda=25, \mu=17 \Rightarrow \alpha=36$

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